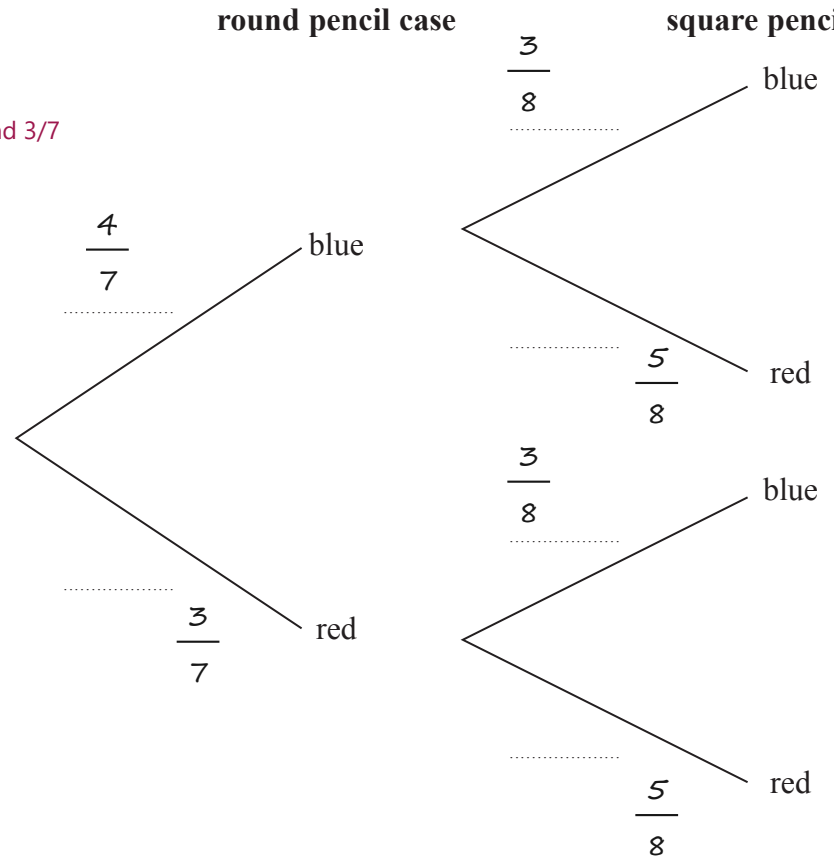
8 Sameena has a round pencil case and a square pencil case.

There are 4 blue pens and 3 red pens in the round pencil case.
There are 3 blue pens and 5 red pens in the square pencil case.

Sameena takes at random one pen out of each pencil case.

(a) Complete the probability tree diagram.



1 mark
Round: 4/7 and 3/7

1 mark
Square: 3/8, 5/8 on both branches

(2)

(b) Work out the probability that the pens Sameena takes are both red.

$$P(\text{both red}) = \frac{3}{7} \times \frac{5}{8} = \frac{15}{56}$$

1 mark
Multiply red-path probabilities

1 mark
15/56

(2)

(Total for Question 8 is 4 marks)